

Jens Laufer

Forward Deployed Engineer — autonomous AI agents and the harness that makes them reliable

Harness & Agentic Engineering · Forward Deployed · Python · LLM agents / MCP · RAG · Enterprise integration · Java / Spring Boot · Vue 3 · Docker

~29 years of software engineering (freelance since 1997), a second track in data science and machine learning — and since late 2025 harness and agentic engineering: the scaffolding around an LLM that turns a language model into a system you can leave running. Complex projects delivered agile and test-driven, always close to the client.

AVAILABLE immediately	WORKS remote only	REMOTE SHARE 100 % 0 % on-site
HOURLY RATE €100/h on-site €95/h remote · excl. VAT	BASED IN Karlstein am Main Germany	

FOCUS

~29 years of software engineering (since 1997), freelance throughout.

Java/Spring Boot microservices with Vue.js and React front-ends — end-to-end with Docker, Kubernetes and GitLab CI.

Strictly test-driven (TDD, JUnit, Mockito, Playwright/Selenium).

Second track in data science and machine learning — Udacity ML Engineer & Data Analyst, Kaggle, writer at Towards Data Science.

Broad industry exposure: banking/finance, insurance, automotive, telecommunications, travel, industry, retail, marketing and startups.

ROLES

- Full-stack engineering
- Solution design
- Backend engineering
- DevOps
- Build & configuration management
- Data analysis
- Prototyping
- Machine learning
- Jenkins plugin development
- Maven plugin development

SKILLS

LANGUAGES

- Java
- JavaScript
- TypeScript
- Python
- R
- Groovy

MICROSERVICES / BACKEND

- Spring Boot
- Spring
- Spring Data
- Spring Integration
- Spring Security
- OAuth2
- Microservices
- Feign
- Sleuth
- Zipkin
- Python Eve
- Python Flask
- Java EE
- EJB

PERSISTENCE / DATABASES

- MySQL
- MongoDB
- PostgreSQL
- Oracle
- Neo4j
- JPA
- Hibernate

DEVOPS / CLOUD

- Docker
- Kubernetes
- Rancher
- GitLab CI
- Jenkins
- Maven
- Grunt
- npm
- Artifactory
- AWS (EC2, S3, SageMaker, VPC)
- Microsoft Azure
- DigitalOcean

TESTING

- JUnit
- TDD
- Mockito
- Selenium
- DBUnit
- Playwright

MACHINE LEARNING

- scikit-learn
- Keras
- DeepLearning4J
- CNN
- Transfer Learning
- Supervised/Unsupervised
- k-means
- PCA
- t-SNE
- Reinforcement Learning
- OpenAI Gym
- Topic Modeling

TOOLING / OTHER

- Git
- GitLab
- Subversion
- Jira
- Confluence
- Kibana
- ANTLR
- JavaCC
- ZXing (QR)
- PayPal API
- RDF/XML/JSON
- SAP (BAPI)
- Phonegap
- JBoss
- Tomcat

FRONTEND

- Vue.js
- React
- AngularJS
- GWT
- Quasar
- HTML
- CSS
- Bootstrap

SECURITY / IAM

- Keycloak
- Spring Security
- OAuth2

API / INTEGRATION

- REST
- Swagger
- OpenAPI
- SOAP
- Webservices
- CXF
- XFire
- Kafka
- ServiceMix

DATA SCIENCE

- pandas
- numpy
- dplyr
- Jupyter
- RMarkdown
- ggplot
- seaborn
- Vega-Lite
- A/B Testing
- Webscraping

PROJECT HISTORY (SELECTION FROM 2009)

since 10/2025

ongoing

Autonomous AI business partner — Harness & agentic engineering

Solytics GmbH · Remote — AI / Automation

Harness engineering · Agent engineering · AI system architecture · DevOps

A self-directing agent built on Claude that plans, writes code, tests, deploys and reports around the clock. Built as a **harness engineer** — the engineering around the model: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **observability** over every run, **budget and rate-limit control** with model routing, plus recovery when a run is killed mid-flight. More than 40 purpose-built checks catch the agent's mistakes before they reach a human. Event-driven over Telegram, GitHub and systemd; runs on own hardware behind a local multi-model proxy.

- Claude / Anthropic API
- LLM-Agenten
- Agent Harness
- Tool Use / MCP
- RAG
- Prompt Engineering
- Python
- systemd
- Telegram Bot API
- Docker
- TDD

07/2022–04/2024

22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry

Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

Java Spring Boot MongoDB JUnit TDD React Docker GitLab GitLab CI Spring Security
Keycloak Mockito OpenAPI Swagger REST Microservices Playwright

01/2022–06/2022

6 months

Payment back-end migration Java EE → Spring Boot

Condor · Remote — Marketing

Backend engineering

Migrated an EJB-based Java EE payment back-end to Spring Boot and shipped it as a Docker container.

Java Java EE JPA Spring Boot MySQL Docker Mockito JUnit TDD

04/2021–11/2021

8 months

Software and data analysis in marketing

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Prototyping · Solution design · Data analysis

Server-side services in Java and Spring Boot: web-scraping microservices plus microsites (Vue.js, Quasar) with a Spring Boot back-end for calculators, surveys and questionnaire funnels; data analysis in R. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Feign), persisted in PostgreSQL and MongoDB. Built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes.

Scrum Spring Boot Docker Kubernetes Microservices REST R Python Java OpenAPI Feign
JUnit TDD Mockito Vue.js Quasar MongoDB dplyr GitLab CI GitLab PostgreSQL
Python Eve Python Flask DigitalOcean Webscraping A/B Testing Topic Modeling

01/2021–03/2021

3 months

Marketing campaign tooling

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Solution design

Tools that generate marketing campaigns and publish them to Google Ads and Twitter Ads: Java/Spring Boot microservices with REST interfaces (OpenAPI, Feign), Vue.js/Quasar front-end, storage in AWS S3. Designed and built in a microservices architecture. Built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes.

Scrum Spring Boot Docker Kubernetes Microservices REST Java OpenAPI Feign JUnit TDD
Mockito Vue.js Quasar AWS S3 GitLab CI GitLab

12/2019–12/2020

13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin; built and maintained the GitLab CI pipelines.



07/2019–12/2019

6 months

Training a stock-trading agent

JiGaoTrading · Remote — Finance

Machine learning · Data analysis

An agent that trades autonomously on Chinese stock exchanges (reinforcement learning).



07/2015–07/2017

25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.



02/2015–06/2015

5 months

Invoice management platform (SME) — debt-collection integration

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Microservice integrating a debt-collection provider, including a single-page admin front-end.



01/2015–02/2015

2 months

Invoice management platform (SME) — OAuth2 service

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

OAuth2 microservice with an administration front-end.



09/2014–12/2014

4 months

Invoice management platform (SME) — prototype

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering · Solution design · Prototyping · Build & configuration management

Evaluation of microservices and SPA approaches, prototype for an invoice management platform.



05/2011–04/2012

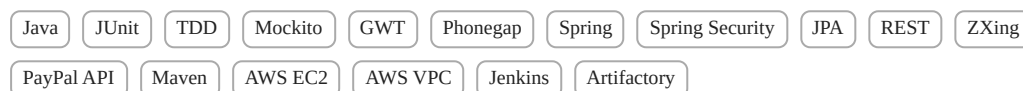
12 months

QR-code based payment solution

Paycoo Ltd. · London — Finance, Startup

Full-stack engineering · Solution design · Build & configuration management

Money transfer between accounts by scanning a QR code; hybrid app (GWT) embedded in native mobile apps for broad platform coverage.



05/2009–03/2010

11 months

Master data migration at a major bank

Commerzbank · Frankfurt a. Main — Banking, Finance

Backend engineering

Workflows for persisting master data, a canonical XML model, RDF description for machine processing.



CONTACT & CREDENTIALS

CONTACT & PROFILES

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LINKEDIN	linkedin.com/in/jenslaufer
KAGGLE	kaggle.com/jenslaufer
PUBLICATION	Towards Data Science — author

EDUCATION

11/1997
Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent)
University of Erlangen-Nuremberg
05/1989
Abitur (university entrance qualification)
Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018
Machine Learning Engineer Nanodegree
Udacity
Verify →
08/2017
Data Analyst Nanodegree
Udacity
Verify →